The IP Multimedia System (IMS)

For many years, Over-The-Top (OTT) services have led to a significant decrease in mobile (and fixed) networks operator's revenue all around the world. Being flexible, quick to innovate and quick to market, many of OTT players have turned into billion-dollar service providers.

Network operators, often reliant and bound to both technical and administrative specifications, face significant challenges. Without adopting to the new environment and often too slow and far behind the drag curve, network operators risk being turned into mere "bit-pipe providers".

The IP Multimedia System – IMS – permits and enhances real-time, multimedia mobile services such as rich voice, video telephony, messaging, conferencing, streaming and many more services by providing a standardized, reusable platform to better address the "develop, deploy and integrate advanced services" for both consumer and enterprise sectors.

Course objectives:

This training aims at providing a technical insight to the IP Multimedia System (IMS), specified by 3GPP.

We will discuss business aspects, concepts and architecture of IMS.

Participants will learn about the role, definition and operation of SIP, SDP and DIAMETER protocols, thus gaining a solid foundation on understanding the many benefits of a modern IMS.

Analysis of selected uses cases and service scenarios, such as IMS Registration procedures, Voice and Video Call Setup, Instant Messaging etc. shall support the understanding of an IMS's the huge potential for network operators.

Duration and Instructional methods:

Option 1: On-Site Training: 3 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 3 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 6 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

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Target Audience:

The training is designed for engineers, managers and network operator's staff who require a solid foundation in IMS to shape successful services.
Service providers and software developers will appreciate the technical background, enabling a closer cooperation with network operators.
It will also benefit to those in the wider technical community who have the need and interest in understanding the benefits and architecture of IMS, its features and capabilities.

Course Contents:

Introduction and Business Aspects

- Drivers for Rich Communication Services
- Business Aspects
- Over-the-top (OTT) services vs. network provided services
- Standardization work

IMS architecture

- Reference Architecture
- Application Layer, Service Control Layer and Media Layer
- Component Description – the IMS building blocks
- IMS Call Servers – CSCF
- Data Bases – HLR and SLF
- Media Resource Function – MRFP and MRFC
- Breakout Gateway – BGCF, MGCF, MGW, SGW
- IMS Reference Points
- Protocols for IMS
- IMS User Identifications and User Profiles

Role and operation of SIP (Session Initiation Protocol)

- SIP Functionality
- SIP Sessions
- UA Registration with SIP
- Message Routing with SIP
- SIP Message Format
- The SIP Session
- Session description with SDP (Session Description Protocol)
- Selected SIP Options and Extensions

Media encoding and transport

- Audio Media and Audio Codecs in IMS
- Video Encoding and Video Codecs in IMS
- Text Encoding in IMS
- Bandwidth requirements for media formats
- Media transport with RTP and RTCP
- Payload Types
- Secure RTP (SRTP)

AAA, Policy and Charging Control in IMS

- AAA functions in IMS
- Role and application of DIAMETER protocol in IMS
- Selected DIAMETER messages and commands
- Policy and Charging Control (PCC) in IMS
- PCC Architecture
- Push and Pull Mode for Policy Control
- Accounting in IMS
- Online and Offline Charging in IMS
- Event-based Charging in IMS

IMS basic procedures and service interworking examples

- IMS Security features
- IMS Registration Procedure
- Audio- and Video Call handling – IMS Session Establishment
- IMS Instant Messaging Service
- IMS Presence Service
- Push-over-Cellular (PoC) Service
- IMS Conferencing Service
- PSTN / ISDN Simulation Services

Pre-requisites:

Attendance to this training assumes a solid understanding of mobile and fixed network architectures and the traditional approaches to deliver services to subscribers.
A basic understanding of SIP and DIAMETER protocols used in Mobile and IP Networks would be an advantage as would be basic understanding of SIP signaling concepts.

Course Code: IMS01